ORGANIC CHEMISTRY FRONTIERS

RESEARCH ARTICLE

Published on 28 October 2024
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Cite this: *Org. Chem. Front.*, 2025, **12**, 243

Magic effect of diphenylprolinol silyl ether for the enantioselective allenation of 2-alkynals†

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Received 1st June 2024,
Accepted 16th October 2024

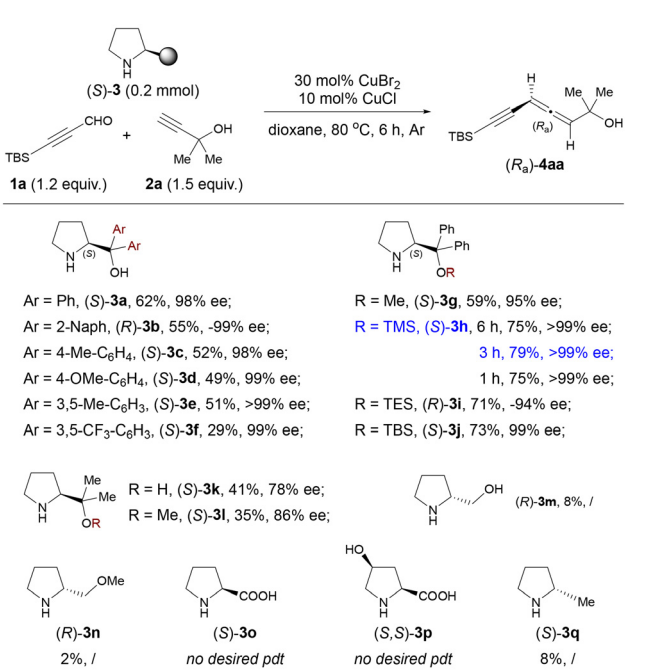
DOI: 10.1039/d4qo00984c

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Diphenylprolinol silyl ethers have been successfully identified as efficient chiral amines in the enantioselective allenation reaction of terminal alkynes and 2-alkynals. This reaction provides a diverse set of chiral allenynes with excellent enantioselectivity and decent yields. Furthermore, the current protocol is also compatible for the late-stage modification of some complex bioactive molecules, highlighting its potential in drug discovery.

Allenynes are among the simplest and the most common structural units in numerous allenic natural products.[1,2] In addition, their unique conjugated structure has attracted more and more attention from chemists interested in molecular materials.[3,4] One of the most useful protocols for the synthesis of allenynes relies on transition-metal-catalyzed cross coupling of terminal alkynes with propargylic alcohol derivatives[5] or allenic halides.[6] However, such reactions in an enantioselective manner have remained a challenge. Recently, Liu reported the asymmetric radical coupling of 1,3-enynes with terminal alkynes to afford optically active tetrasubstituted allenynes.[7] On the other hand, transition-metal-promoted allenation of terminal alkynes (ATA) with aldehydes or ketones has evolved as one of the most powerful approaches for the synthesis of allenes.[8,9] Recently, we have developed the enantioselective allenation of terminal alkynes (EATA) with 2-alkynals in the presence of (*S*)- or (*R*)-α,α-diphenylprolinol, affording a series of synthetically challenging chiral 1,3-disubstituted allenynes.[10] However, the yields were not satisfactory in many cases. Here we wish to report our recent observation that such an EATA reaction may be greatly improved by applying the magic effect of chiral diphenylprolinol silyl ether,[11] which gives access to a wide range of axially chiral allenynes in good yields with excellent enantioselectivity.

Our study began by optimizing the commercially available chiral amine in the reaction of 3-*t*-butyldimethylsilylpropynal **1a** and tertiary propargylic alcohol **2a** in 1,4-dioxane at 80 °C (Scheme 1). (*S*)-α,α-Diphenylprolinol **3a** was tested by combining it with $CuBr_2$ (30 mol%) and CuCl (10 mol%) as catalysts, affording the desired product (R_a)-**4aa** in 62% yield with 98%



Scheme 1 The effect of chiral amines. The reaction was conducted with 0.24 mmol **1a**, 0.3 mmol **2a**, 0.2 mmol (*S*)-**3**, $CuBr_2$ (30 mol%) and CuCl (10 mol%) in 1.8 mL of 1,4-dioxane at 80 °C under an argon atmosphere. The yield was determined by ^1H NMR analysis using CH_2Br_2 as an internal standard. The ee value was determined by chiral HPLC analysis of the isolated product.

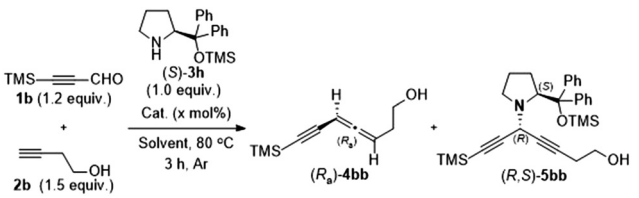
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†Electronic supplementary information (ESI) available. CCDC 2039513. For ESI and crystallographic data in CIF or other electronic format see DOI: https://doi.org/10.1039/d4qo00984c

ee. Attempts to optimize the aryl group at the α-position of prolinol proved unsuccessful (**3b–3f**). To our delight, silyl-protected diphenylprolinols were effective in this reaction: the yield of (R_a)-**4aa** was improved to 79% in 3 h when (*S*)-α,α-diphenylprolinol *O*-TMS ether **3h** was used. Other silyl ether groups (OTES and OTBS) showed similar efficiency. (*S*)-α,α-Dimethylprolinol **3k** and its methyl ether (*S*)-**3l** turned out to be detrimental both for efficiency and enantioselectivity. Other chiral amines, such as L-prolinol and L-proline and its derivatives, were virtually ineffective. (*S*)-2-Methylpyrrolidine **3q** was also examined and afforded the allenyne only in 8% yield.

The reaction of 3-trimethylsilylpropynal **1b** and 3-butyn-1-ol **2b** with (*S*)-α,α-diphenylprolinol **3a** delivered the desired product (R_a)-**4aa** only in 32% yield (Table 1, entry 1).[10] To our delight, the yield was improved to 48% when the reaction was conducted with (*S*)-**3h** for 3 h (Table 1, entry 2). We next conducted extensive investigation on other experimental parameters with (*S*)-**3h** as the optimal chiral amine. Mono-metallic or other mixed metallic couples showed low efficiency and gave the allenyne (R_a)-**4bb** together with the 1,4-diyn-3-yl amine intermediate (*R*,*S*)-**5bb** (Table 1, entries 3–10). Solvent screening revealed that MTBE was the optimal, in which the reaction at 70 °C could give the highest yield of 57% (entries 11–15). This EATA reaction could also proceed on a 1.0 mmol scale under the co-catalysis of CuCl and $CuBr_2$ (Table 1, entry 16).

Table 1 Optimization of the reaction conditions with (S)-**3h**[a]



Entry	Cat. (*x* mol%)	Solvent	(R_a)-**4bb** yield, ee	(*R*,*S*)-**5bb** yield
1[b]	$CuBr_2$ (30), CuCl (10)	Dioxane	32, 98	—
2	$CuBr_2$ (30), CuCl (10)	Dioxane	48, 99	0
3	$CuBr_2$ (30)	Dioxane	33, 98	6
4	CuCl (10)	Dioxane	1, —	80
5	$CuCl_2$ (30), CuCl (10)	Dioxane	40, 99	9
6	$Cu(OAc)_2$ (30), CuCl (10)	Dioxane	1, —	63
7	CuO (30), CuCl (10)	Dioxane	3, —	77
8	$ZnBr_2$ (30), CuCl (10)	Dioxane	21, 96	2
9	$CuBr_2$ (30), CuBr (10)	Dioxane	47, 94	3
10	$CuBr_2$ (30), CuI (10)	Dioxane	47, 90	4
11	$CuBr_2$ (30), CuCl (10)	THF	50, >99	5
12[c]	$CuBr_2$ (30), CuCl (10)	Et_2O	36, >99	17
13[d]	**$CuBr_2$ (30), CuCl (10)**	**MTBE**	**57, 99**	**0**
14	$CuBr_2$ (30), CuCl (10)	CPME	50, 98	0
15	$CuBr_2$ (30), CuCl (10)	Toluene	53, 97	0
16[d,e]	**$CuBr_2$ (30), CuCl (10)**	**MTBE**	**50, 99**	**3**

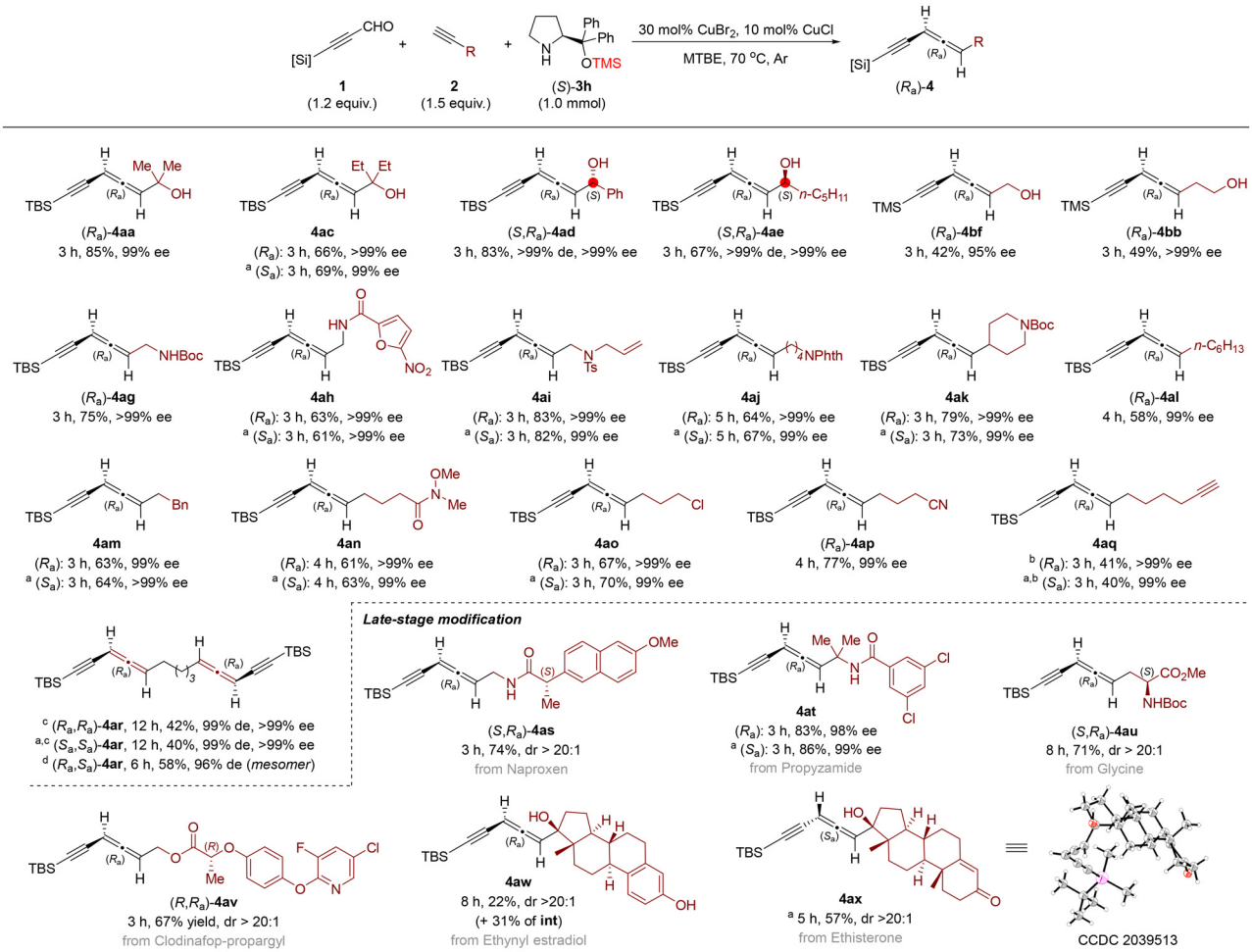
[a] The reaction was conducted with 0.24 mmol **1b**, 0.3 mmol **2b**, 0.2 mmol (*S*)-**3h**, and the indicated amount of catalysts in 1.8 mL of solvent under an argon atmosphere, unless otherwise noted. The yield was determined by ^{1}H NMR analysis using CH_2Br_2 as an internal standard. The ee value was determined by chiral HPLC analysis of the isolated product. [b] The reaction was conducted with (*S*)-**3a** (1.0 mmol) for 6 h. [c] 50 °C. [d] 70 °C. [e] The reaction was conducted with 1.0 mmol (*S*)-**3h** in 5.0 mL of MTBE.

With the optimized conditions in hand, we next investigated the generality of the reaction substrates. First, the scope of terminal alkynes **2** was studied (Scheme 2). Reactions of a wide range of propargylic alcohols (**2a** and **2c–2f**) afforded the corresponding allenynes (**4aa**, **4ac–4ae**, and **4bf**) in good to moderate yields with excellent ee (up to >99% ee). It should be noted that chiral secondary propargylic alcohols **2d** and **2e** also worked under the optimized conditions, affording the corresponding allenynes **4ad** and **4ae** with excellent enantio- and diastereoselectivities. Amine-tethered terminal alkynes **2g–2k** were also used, furnishing the corresponding products **4ag–4ak** in good yields and high ee. Moreover, alkyl-substituted terminal alkynes also afforded the desired products **4al** and **4am** in decent yields with excellent ee. An attractive range of synthetically useful functional groups such as alkenyl (**4ai**), Weinreb amide (**4an**), Cl (**4ao**), and CN (**4ap**) were compatible and afforded in good yields with excellent ee. It is worth noting that 1,7-octadiyne **2q** was also tolerated, affording mono-allenyne **4aq** in decent yield with high ee. Interestingly, when the ratio of **1a** : **2q** : **3h** was adjusted to 2.4 : 1 : 2.4, bis-allenyne **4ar** could be isolated with good enantio- and diastereoselectivities (see Scheme S1 in the ESI† for the comparison between (*S*)-**3a** and (*S*)-**3h** in the EATA reaction).

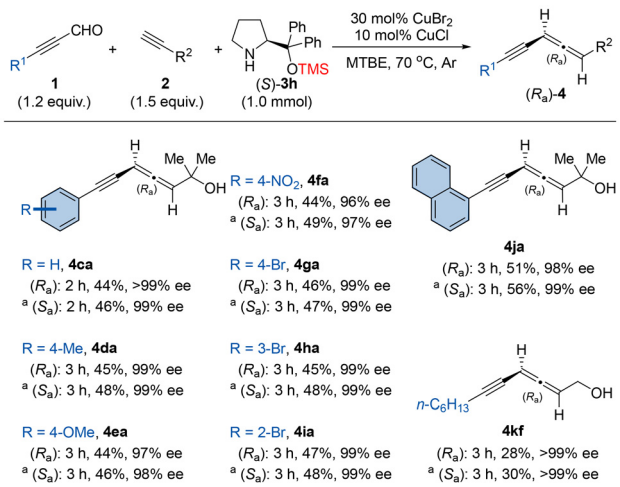
The high functional group compatibility and broad substrate scope of this method encouraged us to examine the late-stage modification of a series of biologically relevant molecules containing terminal alkynes (Scheme 2). The non-steroidal anti-inflammatory drug naproxen may also be incorporated into the terminal alkyne, giving optically active allenyne (*S*,R_a)-**4as** in 74% yield. Both enantiomers (R_a)- and (S_a)-**4at** derived from propyzamide were obtained in high yields with excellent ee when (*S*)- and (*R*)-**3h** were used. Glycine-tethered terminal alkyne (*S*)-**2u** afforded the corresponding product (*S*,R_a)-**4au** in 71% yield. More structurally complex biologically active molecules, including clodinafop-propargyl (**2v**), ethynyl estradiol (**2w**) and ethisterone (**2x**), were studied and the corresponding functionalized products were afforded in good yields. The absolute configuration of the allenyne unit was established by X-ray crystallography analysis of **4ax**.[12]

Finally, the scope of 2-alkynal substrates was also assessed under the optimal conditions. In addition to 3-silylpropynals, other aryl substituted 2-alkynal substrates were also suitable for this EATA reaction. An array of 3-phenylpropynal derivatives bearing both electron-donating (**1d** and **1e**) and electron-withdrawing (**1f** and **1g**) groups on the benzene ring delivered the corresponding products with excellent enantioselectivities and decent yields (44%–49%). There is no obvious steric effect since 3-aryl-2-alkynals with a bromine atom at the *para*, *ortho*, or *meta*-position of the benzene ring, all afforded the allenynes **4ga–4ia** in similar yields with high ees. 3-(1-Naphthalenyl)propynal **1j** was also compatible, affording product **4ja** with excellent enantioselectivity. Even the reaction of 3-alkyl substituted 2-alkynal **1k** afforded both enantiomers of **4kf** in acceptable yields with >99% ee (Scheme 3).

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Scheme 2 Scope of terminal alkynes and late-stage modification. The reaction was conducted with 1.2 mmol **1**, 1.5 mmol **2**, 1.0 mmol (*S*)-**3h**, 30 mol% of $CuBr_2$, and 10 mol% of CuCl in 5.0 mL of MTBE at 70 °C under an argon atmosphere, unless otherwise noted. [a] (*R*)-**3h** was used instead of (*S*)-**3h**. [b] 2.0 mmol **2q** was used. [c] The reaction was conducted with 2.4 mmol **1a**, 1.0 mmol **2q**, 2.4 mmol (*S*)-**3h**, 60 mol% of $CuBr_2$, and 20 mol% of CuCl in 10.0 mL of MTBE at 70 °C under an argon atmosphere. [d] The reaction was conducted with 0.6 mmol **1a**, 0.75 mmol (R_a)-**4aq**, 0.5 mmol (*R*)-**3h**, 30 mol% of $CuBr_2$, and 10 mol% of CuCl in 2.5 mL of MTBE at 70 °C under an argon atmosphere. See details in the ESI.†



Scheme 3 Scope of 2-alkynals. The reaction was conducted with 1.2 mmol **1**, 1.5 mmol **2**, 1.0 mmol (*S*)-**3h**, 30 mol% of $CuBr_2$, and 10 mol% of CuCl in 5.0 mL of MTBE at 70 °C under an argon atmosphere. [a] (*R*)-**3h** was used instead of (*S*)-**3h**. See details in the ESI.†

A plausible mechanism has been proposed (Fig. 1a):[10] Initially, 2-alkynal **1** reacts with proline (*S*)-**3** and terminal alkyne **2** to afford 1,4-diyn-3-yl amine (*R*,*S*)-**5** *via* an A^3-coupling reaction. Then intermediate (*R*,*S*)-**5** undergoes sequential regio- and stereoselective 1,5-H transfer and *trans* β-*N*-elimination to yield the chiral allenyne (R_a)-**4** in the presence of $CuBr_2$. According to DFT calculations,[10] the 1,5-H transfer is the rate-limiting step of the whole process.

To further elucidate the remarkable differences of the efficiency between diphenylprolinol **3a** and **3h**, DFT calculations were performed to unveil the influence of the O-TMS protecting group on the reactivity of the reaction (Fig. 1b): for the 1,5-H transfer of the methylene hydrogen (H^a) to the hydroxyalkyl-substituted triple bond, **TS1_a_TMS** exhibits a free energy barrier of only 11.6 kcal mol^{-1} which is 13.2 kcal mol^{-1} lower than that of **TS1_a** (24.8 kcal mol^{-1}).

Careful examination of the geometries of **Int1** reveals that it is stabilized by the coordination of copper with both the nitro-

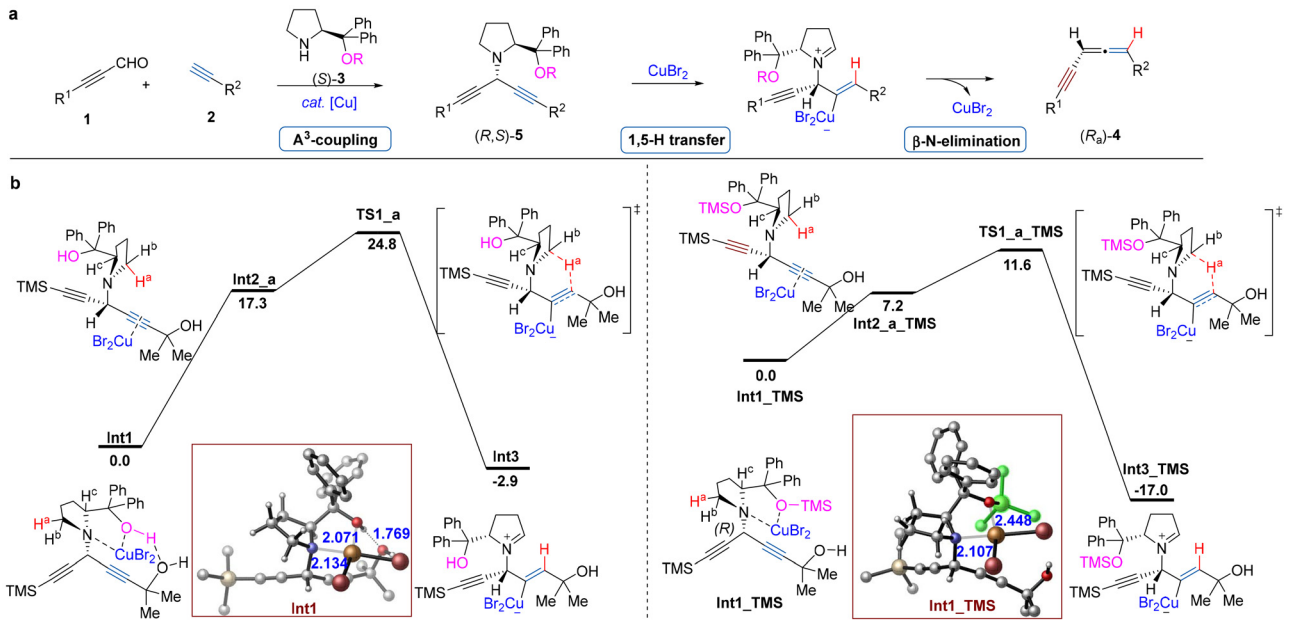


Fig. 1 (a) The reaction mechanism. (b) DFT calculations on the 1,5-H transfer step from **Int1** to **Int1_TMS** (ΔG in kcal mol^{-1} with respect to **Int1** or **Int1_TMS**. Bond lengths are given in Angstroms).

gen atom and the hydroxyl oxygen atom with distances of 2.134 and 2.071 Å, respectively, as well as hydrogen bonding between the prolinol hydroxyl hydrogen and the propargylic hydroxyalkyl oxygen (1.769 Å). The subsequent isomerization of **Int1** to **Int2_a** involves the dissociation of copper from the nitrogen and oxygen atoms, coupled with its coordination to the carbon–carbon triple bond, resulting in an endergonic process of 17.3 kcal mol^{-1}. The high thermodynamics stability of **Int1** results in a low reactivity for the [1,5]-hydride migration. In comparison, no hydrogen bonding exists in **Int1_TMS**, and the O-TMS protector weakens the coordination of copper with the oxygen atom with a bond length of 2.448 Å, which is 0.377 Å longer than that in **Int1**. The transformation of **Int1_TMS** to **Int2_a_TMS** requires only 7.2 kcal mol^{-1}, suggesting a relatively lower stability of **Int1_TMS** compared to that of **Int1**. Therefore, the O-TMS unit on diphenylprolinol facilitates the 1,5-H transfer process primarily by destabilizing **Int1_TMS**.

Conclusions

In summary, we have developed the enantioselective allenation reaction of 2-alkynals in the presence of (*S*)- or (*R*)-α,α-diphenylprolinol silyl ether. This reaction delivers a diverse set of chiral allenynes featuring high efficiency and a broad substrate scope. The current protocol may also be applied to the late-stage modification of some bioactive molecules, highlighting its potential in drug discovery. Further applications of this protocol including the enantioselective total synthesis of natural products containing allenynes are being actively pursued in our laboratory.

Author contributions

S. M. and H. Q. conceived and directed the project and designed the experiments. G. W. conducted most of the experiments. S. F. and W. L. participated in the substrate scope investigation. X. Z. and Z. L. performed the density functional theory calculations. G. W, X. Z., H. Q. and S. M. checked the experimental data and wrote the manuscript.

Data availability

All detailed procedures, characterization data, and spectra are available in the ESI.† Crystallographic data for **4ax** have been deposited at the Cambridge Crystallographic Data Center under the deposition number CCDC 2039513.†

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

Financial support from the National Key R&D Program of China (Grant No. 2022YFA1503200) is greatly appreciated. We thank Dr Haibo Xu in this group for reproducing the results of (R_a)-**4aa** and (S_a)-**4ai** in Scheme 2 and (R_a)-**4ca** in Scheme 3.

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12 CCDC 2039513† (**4ax**) contains the supplementary crystallographic data for this paper.